

Question Paper

Exam Date & Time: 14-Nov-2019 (10:00 AM - 01:00 PM)



CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

I Semester of BSc Biological Sciences Examination November 2019

Date: 14/11/2019 Day: Thursday Time: 10.00 am to 1.00 pm

CHEMISTRY [BS109]

Marks: 70

Duration: 180 mins.

MCQ

Answer all the questions.

Section Duration: 30 mins

Choose the correct answer.

- 1) The mathematical equation show that, how rate of reaction change with concentration of reactant is called (1)
[Differential rate law](#) [Rate constant](#) [Order of reaction](#) [Molecularity](#)
- 2) Which repulsion between electron-pairs is strongest one? (1)
[Lone pair - Lone pair](#) [lone pair - bond pair](#) [Bond pair - bond pair](#) [None of these](#)
- 3) What is unit the for rate constant for second order reaction? (1)
[Mol.⁻¹dm⁻³s⁻¹](#) [Mol.²dm⁻²s⁻¹](#) [Mol.⁻¹dm³s⁻¹](#) [Mol.⁻²dm⁻²s⁻¹](#)
- 4) If the rate of reaction becomes 2 fold when temperature increasing from 37 °C to 47 °C then what is the value of activation energy? (1)
[13.66 K Cal](#) [13.66 Cal](#) [18.66 K Cal](#) [18.66 Cal](#)
- 5) Which is the more fundamental property of the elements? (1)
[Atomic weight](#) [Ionic radius](#) [Atomic number](#) [Ionization energy](#)
- 6) What is the dimension of force? (1)
[kg. m. sec⁻²](#) [kg. m. sec](#) [kg.m⁻¹.sec⁻²](#) [kg.m.sec⁻¹](#)
- 7) How many lone pairs of electrons are present on nitrogen atom of ammonia molecule? (1)
[0](#) [1](#) [2](#) [3](#)
- 8) The equation for half-life time for second order reaction is (1)
[t_{1/2} = ln2 / Co](#) [t_{1/2} = ln2 × k](#) [t_{1/2} = 1 / kCo](#) [t_{1/2} = ln2 + k](#)
- 9) Value of bond angle in CH₄ _____ (1)
[90° & 120°](#) [90° & 120°](#) [109°28'](#) [90°](#)
- 10) During the process if mass and energy both are constant then system is _____. (1)
[Isolated](#) [Adiabatic](#) [Closed](#) [Open](#)
- 11) Find the **slope** (m) and **intercept** (C) of the following equation 2y + 14x = 8 (1)
[m = -4, c = 7](#) [m = -7, c = 4](#) [m = 2, c = 4](#) [m = 4, c = -2](#)
- 12) Form the following organic molecules, which molecule shows inductive effects? (1)
[Ethane](#) [But - 2 - ene](#) [Chloroethane](#) [1,3, - butadiene](#)
- 13) Criteria for reaction to feasible is _____. (1)
[ΔG = 0](#) [ΔG > 0](#) [ΔG < 0](#) [None of this](#)

- 14) Which pairs follows isoelectronic principle? (1)
- [NO⁻ & NO²⁺](#) [BF₄ & CH₄](#) [CO₂ & SO₃](#) [None of these](#)
- 15) The stereoisomer which is non-super imposable on its mirror image is known as _____. (1)
- [Tautomers](#) [Enantiomers](#) [Homomers](#) [Diastereomers](#)
- 16) The configuration of carbon atom in excited state is _____. (1)
- [1s² 2s² 2p²](#) [1s² 2s¹ 2p⁴](#) [1s² 2s² 2p³](#) [1s² 2s¹ 2p³](#)
- 17) A tetrahedral atom attach to four different substituent is known as _____. (1)
- [Asymmetric center](#) [Chirogenic center](#) [Chiral center](#) [All of these](#)
- 18) The shape of sp² hybridization is _____. (1)
- [Tetrahedral](#) [Linear](#) [Complex](#) [Trigonal](#)
- 19) According to VSEPR theory, the shape of the NH₃ is _____. (1)
- [Tetrahedral](#) [Square planar](#) [Distorted pyramidal](#) [Trigonal pyramidal](#)
- 20) The self-linking property of carbon is called _____. (1)
- 1.
- [Polymerization](#) [Oligomerization](#) [Catenation](#) [None of above](#)

Section I

Answer all the questions.

- 21) What is heat capacity? Obtain the relation for heat capacity at constant pressure and at constant volume. (4)
- [OR]
22) Derive the equation for integration rate law for second order reaction and discuss its characteristics. (4)
- 23) For a first order reaction, the value of rate constant is $8.2 \times 10^{-4} \text{ min}^{-1}$ at 45°C. Calculate the time for the reaction to decrease the concentration from 0.02 M to 0.006 M and time for 70% reaction completion. (4)
- 24) Write the difference between order of reaction and molecularity. (2)
- 25) Show that internal energy is state function. (3)
- [OR]
26) Discuss the properties of enantiomers and diastereomers. (3)
- 27) The rate constant for one reaction is $1 \times 10^{-4} \text{ min}^{-1}$ at 27°C and $2 \times 10^{-4} \text{ min}^{-1}$ at 37°C. Calculate the activation energy of the reaction and rate constant at 57 °C. (4)
- 28) Obtain the relation work and kinetic energy. (3)

Section II

Answer all the questions.

- 29) What is Isomers? Give the class of Isomers. (2)
- 30) Explain "All chirogenic center are stereogenic center but reverse is not true". (2)
- 31) What is Octet? Explain the Octet rule? (3)
- [OR]
32) What is systems? Discuss the types of systems. (3)

- 33) Discuss the defects of Mendeleev's periodic table. (3)
- 34) Construct the molecular orbital diagram of N_2 ($N = 7$) molecule and calculate the bond order. (4)
- 35) Explain "'SP' Hybridization in detail. (3)
- [OR]
36) " O_2 is paramagnetic while O_2^{2-} is diamagnetic"(Given: Atomic Number of Oxygen is 8) Explain on the basis of Molecular Orbital Theory. (3)
- 37) Explain the Lowery Bronsted concept of acid and base by giving suitable examples. (3)
- 38) Explain ionization energy and discuss the factors affecting the magnitude of ionization energy. (4)
- 39) Define the term: (1) Modern periodic law (2) Ionization energy Hybridization (3)
- [OR]
40) Explain BF_3 and BF_4^- ion using VSEPR theory. (3)
- 41) Explain the resonance effect with suitable example. (3)

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